News-Based Inflation Risk and Price Dynamics in Africa: Evidence from a Panel VAR

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Abstract

Inflation surveillance in Africa is constrained by reporting lags and sparse high-frequency statistics. We construct the African Inflation Risk Index (AIRI), a rule-based, fully traceable measure of inflation risk extracted from domestic news corpora for 46 African economies over 2015–2025. Articles are screened for economic relevance, assigned an ordinal severity tier and a confidence weight, and aggregated to country-month level normalised by total corpus size, so larger media sectors do not mechanically register higher risk; each index value decomposes to the articles and matched terms that produced it. Using a Mean Group panel VAR with heterogeneous Granger causality tests, we find the first lag of AIRI predicts next-period inflation after conditioning on inflation persistence, exchange rate depreciation, money growth and output growth. The relationship is strongly state-dependent: concentrated in high-inflation economies and the 2022–23 global inflation episode, and weak in low-inflation regimes and quieter periods, consistent with attention-dependent expectation formation. Clark–West tests detect incremental predictive content at all horizons, after adjusting for data outliers from South Sudan, though realised forecast gains over macroeconomic benchmarks are marginal. AIRI offers a transparent, near-zero-cost signal that is most informative where conventional data are weakest and inflation risk highest, complementing rather than substituting for official statistics.

Keywords: Inflation forecasting; News-based indicators; Panel VAR; Africa
JEL: E31, E37, C33, O55

1 Introduction

Inflation prediction has been central to macroeconomic management over the years, and the accuracy of the methods employed is key to the strength of policy intervention. Yet approaches to investigating the drivers of inflation have mostly relied on backward-looking data produced with a lag (Orphanides 2001) denying policymakers the ability to act in time. The challenge is particularly pronounced in Africa, where data infrastructure is weaker (Jerven 2013) and institutional limitations make these needs difficult to address.

There is a growing body of literature on the use of high-frequency data, such as online prices and search activities, to track economic conditions before official statistics are released (Choi and Varian 2012; Cavallo and Rigobon 2016). Textual data drawn from news media and other published documents is another source that can be used to analyse current economic conditions. Beyond timeliness, media narratives are themselves economically consequential; Shiller (2017) argues that media shape collective expectations.

Because high-frequency data are large and high-dimensional, machine learning methods have been widely explored to extract predictive signals from them. This approach to economic analyses has shown mixed success: Nakamura (2005) found neural networks outperformed autoregressive benchmarks for US CPI at short horizons, while Choudhary and Haider (2012) found that the performance varied across countries. The deeper limitation is interpretability: these models are characterised as black boxes because, although they capture complex nonlinearities, they do not reveal what drives their predictions (Araujo and Gaglianone 2023). For institutions that must justify decisions publicly, a signal that cannot be explained cannot easily be acted upon.

Rule-based text methods offer a useful alternative to address these gaps. This approach ensures traceability and interpretability, as demonstrated by Baker, Bloom, and Davis (2016) in the Economic Policy Uncertainty index, which counts newspaper articles containing economic uncertainty and policy terms based on specified keywords in dictionaries. The approach has since been extended to GDP (Elshendy and Fronzetti Colladon 2017) and to social media measures tracking realised inflation and survey expectations (Angelico et al. 2022).

This literature nonetheless remains concentrated in advanced economies (Baker, Bloom, and Davis 2016; Ghirelli, Pérez, and Urtasun 2019). The World Uncertainty Index (Ahir, Bloom, and Furceri 2022) relies on standardised international reports rather than domestic corpora. The closest continental attempt (Salisu, Salisu, and Salisu 2024) is single-country. No measure provides inflation-specific risk coverage across African economies, and no study has tested whether such signals predict realised inflation on the continent using panel methods. The gap matters because African economies exhibit exactly the conditions under which narrative signals should be most valuable: high inflation volatility,

rapid exchange rate pass-through, weak expectation anchoring, and severe constraints on official statistics.

We address this gap in two steps. First, we construct the African Inflation Risk Index (AIRI), a news-based measure designed to be traceable, monotonic and comparable across countries with differing media volumes. Articles are screened through a dictionary-based economic relevance filter following lexicon-based approaches to economic and financial text (Shapiro, Sudhof, and Wilson 2022), assigned an ordinal severity tier according to how strongly an issue is framed as threatening, and weighted by signal credibility. Scores are summed over the inflation-tagged subset and divided by total article count for the country-period, so larger media sectors do not mechanically register higher risk. Because construction is rule-based rather than estimated, every value decomposes to the articles and terms producing it, directly addressing the black-box limitation above. Second, we test the index across 46 African countries over 2015–2025 using a Mean Group panel VAR, heterogeneous panel Granger causality tests, and out-of-sample forecast comparisons against autoregressive and macroeconomic benchmarks.

Three findings emerge. The first lag of AIRI predicts next-period inflation after conditioning on inflation persistence, exchange rate depreciation, money growth and output growth, and non-causality is rejected for the panel. The relationship is strongly state-dependent. It is concentrated in high-inflation economies and the 2022–23 inflation episode, and weak or undetectable in low-inflation regimes and quieter periods, a pattern consistent with attention-dependent expectation formation. Forecast evidence is more measured: tests appropriate to nested models detect incremental predictive content up to six months ahead, but realised error improvements over the macroeconomic benchmark are marginal. AIRI is therefore a transparent, near-costless complement to official statistics, most informative where conventional data are weakest and inflation risk highest.

The remainder of the paper proceeds as follows. Section 2 reviews the literature. Section 3 describes the data and the construction of the AIRI, the panel VAR specification, and the forecast evaluation framework. Section 4 presents results. Section 5 concludes with policy implications.

2 Literature Review

2.1 Expectations and the Limits of Realised Price Indices

Conventional inflation measurement relies on realised price indices, the Consumer Price Index and related measures, which record price changes only after they have occurred. This backward-looking character creates a structural information gap: the expectations

driving pricing decisions are invisible to statistical agencies until they have materialised in observed transactions (Mishkin 2007). Knight (1921) and Keynes (1936) established that economic decisions are governed by forward-looking beliefs rather than realised outcomes, and Friedman (1968) and Phelps (1968) formalised this by anchoring inflation in expectations, a dimension price index cannot capture.

The empirical consequences are well documented. Mankiw and Reis (2002) and Sims (2003) showed that information disseminates slowly across agents, generating systematic gaps between actual and expected inflation. Coibion and Gorodnichenko (2012, 2015) demonstrated that these gaps are predictable and persistent, while Coibion et al. (2023) provided experimental evidence that inflation expectations independently alter household behaviour even absent contemporaneous price changes. Weber et al. (2022) confirmed that subjective beliefs drive firm pricing strategies, establishing that the expectation-formation process causally precedes the price dynamics that indices eventually record. Forward-looking inflation surveillance therefore requires access to the information environment in which expectations form, not merely the price records that result.

2.2 News-Based Measurement as a Forward-Looking Alternative

Shiller (2017) identified news media as the primary channel through which economically consequential narratives propagate and shape collective expectations. Tetlock (2007), García (2013), and Nyman, Kapadia, and Tuckett (2021) confirmed empirically that news-based signals contain forward-looking information about market conditions, economic activity, and systemic risk. Baker, Bloom, and Davis (2016) operationalised this insight through the Economic Policy Uncertainty (EPU) index, showing that newspaper-based indicators are strongly associated with investment, employment, and output, and Angelico et al. (2022) extended the approach to social media, constructing real-time measures that track both realised inflation and survey-based expectations.

Applications to inflation specifically remain concentrated in advanced economies. Shapiro, Sudhof, and Wilson (2022) established the link between media narratives and price dynamics in the United States. In Norway, Larsen, Thorsrud, and Zhulanova (2021) used machine learning tools to demonstrate that media-reported topics are strong predictors of inflation. In Europe, Barbaglia, Consoli, and Manzan (2023) found that topic-specific sentiment indicators targeting inflation-related narratives substantially outperform both generic sentiment measures and standard AR benchmarks in out-of-sample forecasting, with predictive gains concentrated at horizons of three to six months. Their finding that topic-specific pillars outperform aggregate uncertainty measures motivates the pillar-level rather than aggregate architecture adopted here, and their horizon results inform the fore-

cast horizons evaluated in Section 4.

2.3 Attention and the Conditional Value of Narrative Signals

Theory does not predict that narrative signals should be uniformly informative, and this distinction is central to the present study. Sims (2003) and Mankiw and Reis (2002) imply that agents economise on attention when the cost of inattention is low, and a subsequent literature establishes that attention to inflation is itself state-dependent. Coibion and Gorodnichenko (2015) show that forecast errors and expectation updating vary systematically with the inflation environment, and household-level evidence indicates that attention to prices rises sharply once inflation becomes salient (Weber et al. 2022; Korenok, Munro, and Chen 2022). Bracha and Tang (2025) similarly document that inflation attention responds nonlinearly to the level of inflation itself.

The implication for news-based measurement is direct. If narratives influence expectations through the attention channel, a media-derived inflation risk signal should carry predictive content precisely when inflation is high or volatile enough to command attention, and should weaken or disappear when it is not. Regime dependence is therefore not an anomaly to be explained after the fact but a prediction the theory generates in advance, and it is a prediction this paper evaluates explicitly by comparing predictive content across country groups defined by inflation history and across the 2022 to 2023 global inflation episode.

2.4 Inflation Dynamics and Data Constraints in Africa

Inflation dynamics in Africa differ from those in industrialised economies in ways that bear directly on measurement. Where demand-side pressures dominate in advanced economies, African inflation is substantially more exposed to supply-side shocks, principally large currency depreciations, global food supply disruptions, and climate-driven volatility in domestic agricultural output (Durevall and Sjö 2012). Inflation mechanisms and persistence also vary widely across the continent, particularly between resource-rich and non-resource-rich economies (Kelikume 2018; Phiri 2016), which implies that empirical work imposing common dynamics across countries risks averaging away the variation of interest.

These characteristics coexist with binding data constraints. Advanced economies increasingly draw on high-frequency administrative sources, whereas African monetary authorities remain largely dependent on the conventional Consumer Price Index (CPI), which is low frequency, subject to substantial publication delay, and insufficiently granular to capture the localised character of price shocks (Blinder et al. 2024; Yeboah, Agyei, and

Korsah 2025). Long-run series are further limited in coverage and comparability. The combined effect is an environment in which conventional inflation-targeting procedures are difficult to operate and real-time policy response is constrained.

These constraints have motivated interest in higher-frequency digital alternatives. Text-mining and natural language processing methods, ranging from lexicon-based scoring to transformer models such as InflaBERT, have delivered nowcasting gains elsewhere using news and social media text (Allard, Teiletche, and Zinebi 2024; Nguyen et al. 2025), and the tone of central bank communication has been shown to affect inflation expectations in African settings (Odu, Umaru, and Ado 2024). Application within the African empirical literature nonetheless remains limited, with non-textual econometric approaches continuing to predominate (Yeboah, Agyei, and Korsah 2025). Existing continental work relies principally on official backward-looking data: Nguyen et al. (2017) employ a Global VAR to identify the drivers of inflation in Sub-Saharan Africa, and Alper, Hobdari, and Uppal (2016) investigate the heterogeneity of inflation dynamics across income groups in the region. The closest news-based attempt, Salisu, Salisu, and Salisu (2024), constructs an economic policy uncertainty index for Nigeria from newspaper text, but is confined to a single country.

2.5 Panel Econometric Methods for Heterogeneous Macroeconomic Panels

Extending news-based inflation analysis from single economies to a continent-wide panel raises estimation issues the existing literature has not had to confront. The heterogeneity documented in Section 2.4 is the first. Pesaran and Smith (1995) show that pooled estimators applied to dynamic panels with heterogeneous slopes yield inconsistent estimates of average effects, since the bias does not vanish as the panel grows. Where dynamics plausibly differ across units, the Mean Group estimator, which estimates country-specific models and averages the coefficients, is the appropriate alternative. Panel VAR estimation itself follows the framework developed by Love and Zicchino (2006) and implemented by Abrigo and Love (2016).

Testing for predictive precedence across heterogeneous units requires comparable care. The pooled Granger causality test imposes a common null that is rarely appropriate in macroeconomic panels, and Dumitrescu and Hurlin (2012) develop a procedure that accommodates heterogeneous causal structures across units. Cross-sectional dependence poses a further threat: African economies are jointly exposed to global commodity prices, dollar funding conditions, and common shocks, and ignoring this dependence invalidates standard panel inference. Pesaran (2006, 2015) provides both a diagnostic test and the common correlated effects approach to estimation under such dependence.

Forecast evaluation raises a final and frequently misunderstood issue. Clark and West (2007) establish that conventional equal-predictive-accuracy tests are misspecified when the competing models are nested, as they are whenever a candidate predictor is added to an autoregressive benchmark, because the additional parameter introduces estimation noise under the null. The adjusted statistic they propose tests for incremental predictive information rather than realised improvement in forecast accuracy, and the two should not be conflated when reporting results. Section 3 sets out how each of these procedures is implemented.

2.6 The Gap

Three limitations therefore characterise the existing literature as it bears on Africa. First, news-based inflation measurement remains concentrated in advanced economies: the EPU index covers no African countries, and the World Uncertainty Index (Ahir, Bloom, and Furceri 2022) is built from standardised international reports rather than domestic corpora, so it captures how African economies are described externally rather than the narrative environment in which domestic agents form expectations. Second, no existing index provides a pillar-level inflation risk measure with continent-wide coverage, and no study has tested whether such signals precede realised inflation across African economies using panel methods capable of accommodating heterogeneous dynamics. Third, the attention literature implies that the value of narrative signals should be conditional on the inflation regime, yet this prediction has not been tested in any setting where inflation regimes vary as widely as they do across the continent.

This gap is consequential because African economies exhibit precisely the conditions under which narrative signals should be most valuable: structural data delays, commodity-driven inflation, rapid exchange rate pass-through, weak expectation anchoring, and limited monetary policy credibility (Buell et al. 2021; Olamide, Ogujiuba, and Maredza 2022; Kasekende, Atingi-Ego, and Durevall 2018). The present study addresses it by subjecting the AIRI inflation pillar to panel econometric testing across 46 African economies, assessing whether the index contains incremental predictive information for realised inflation, whether that predictive content varies systematically across country groups defined by inflation history, and whether it improves the forecast performance of conventional macroeconomic models.

3 Methodology

This section attempts to formalise the evaluation of the predictive power of the AIRI inflation pillar. The Panel VAR is used to establish the statistical relationship between

the lagged values of the AIRI and realised inflation and evaluate its predictive power across varying macroeconomic contexts (high and low inflation). The forecast evaluation checks if the AIRI inflation pillar improves forecast accuracy in models.

3.1 Data

We compute inflation risk as a decomposed component of the aggregate news-based index for Africa. The index captures inflation threats framed in newspaper narrative. The article-level scoring architecture used to derive the index includes economic relevance screening, ordinal severity tiering, confidence weighting, and multi-pillar mapping that identifies the inflation risk. By definition, the inflation risk index used in this paper is the normalised sum of article risk scores restricted to inflation-tagged articles, scaled to policy-friendly levels. The key characteristics of the index include monotonicity, non-negativity, and traceability, i.e. each contribution can be traced to a specific article, its matched dictionary term, severity tier, and weight component. Also, because the denominator is the total corpus size rather than the count of inflation articles, a country with more media output does not mechanically record more inflation risk, hence achieving volume comparability across countries.

3.1.1 Data Processing

Corpus Assembly. All news articles are scraped for each country c at time period t, and the total count of all articles, $N_{c,t} = |I_{c,t}|$, is used as the denominator, where $I_{c,t}$ denotes the set of all articles retained for country c in period t. This allows for the anchoring of the pillar to the full corpus rather than just inflation-related articles, to ensure that the measure reflects the prominence of inflation narrative within the entire media landscape and allows comparison across countries with differing publication volumes.

Economic relevance screening.

$$R_i = 1\{x_i \text{ matches } \texttt{ECON_RELEVANCE_TERMS}\}.$$

$R_i \in \{0, 1\}$. Only relevant articles proceed to scoring.

Every article scraped is passed through a dictionary-based economic relevance filter. An article is tagged relevant if its concatenated title and body matches at least one term in the economic relevance dictionary ($R_i = 1$); otherwise, it is assigned zero. This ensures that other news cycles, such as politics, sports, or crime, will not contaminate the inflation signal.

Inflation risk mapping. Broadly, every relevant article is checked against the full set of risk pillar dictionaries defined in the AERI framework. Let $p \in \{1, \ldots, P\}$ index the P risk pillars tracked by the framework (inflation risk, FX/external risk, debt/fiscal risk, and others), and let $I_{p,i} \in \{0, 1\}$ indicate whether article i matches pillar p's dictionary. This assignment can be multi-label: a single article may simultaneously activate more than one risk pillar. Of the P pillars available, this paper isolates one: inflation risk.

Each article is checked against the inflation risk dictionary, with terms including consumer prices, cost-of-living pressure, price surges, currency-driven pass-through, and others. The indicator $I_{\text{inf},i}$ equals one when the article matches this dictionary and zero otherwise.

$$I_{\text{inf},i} = 1\{x_i \text{ matches } \texttt{INFLATION_PILLAR_TERMS}\}$$

Severity tiering. Articles receive an ordinal severity tier S_i from 0 to 4 based on their relevance, reflecting how strongly the economic issue is framed as threatening. Dictionaries are tiered, escalating from mild concerns to extreme events through deterioration and crisis language. Non-relevant articles are fixed to zero severity.

$$S_i \in \{0, 1, 2, 3, 4\}, \quad S_i = \begin{cases} 0 & \text{if } R_i = 0, \\ \max\{\text{matched tier in } \texttt{SEVERITY_TIER_TERMS}\} & \text{otherwise.} \end{cases}$$

Confidence weighting. A confidence weight W_i is applied to reflect each article's impact according to the credibility of its signal. The following features add to a baseline of one: a high-salience risk cue in the headline adds 0.5, because titles summarise editorial emphasis; body text exceeding a minimum length threshold adds 0.2, to capture substantive content; and activation of at least two pillars adds 0.3, indicating interconnected rather than isolated stress. The weight is capped at the interval $[1, 2]$ so that no single unusually phrased article can dominate the aggregate.

$$W_i = \text{clip}\left(1 + 0.5 \cdot \text{TitleHint}_i + 0.2 \cdot \text{LongText}_i + 0.3 \cdot \text{MultiPillar}_i,\ 1,\ 2\right)$$

where TitleHint_i, LongText_i, $\text{MultiPillar}_i \in \{0, 1\}$ are indicators for a headline risk cue, body length above threshold, and activation of at least two pillars, respectively, with

$$\text{MultiPillar}_i = 1\left\{\sum_{p=1}^{P} I_{p,i} \geq 2\right\}$$

Article-level risk score. The article is scored by multiplying its relevance, severity, and confidence, which ensures logical consistency. Irrelevant articles contribute zero, and mildly framed relevant articles contribute little. Every score rises based on its severity and confidence. Because of this feature, every generated index is traceable to observed textual features.

$$A_i = R_i \cdot S_i \cdot W_i$$

so that $A_i \in \{0\} \cup [1, 8]$: zero when irrelevant, and between 1 (Tier-1 severity, baseline weight) and 8 (Tier-4 severity, maximum weight 2) when relevant.

Aggregation to the country-period inflation risk index.

$$\text{Inflation}_{c,t} = 100 \cdot \frac{\sum_{i \in \mathcal{A}(c,t)} A_i \cdot I_{\text{inf},i}}{N_{c,t}}$$

Substituting the article score gives the fully expanded form:

$$\text{Inflation}_{c,t} = \frac{100}{N_{c,t}} \sum_{i \in \mathcal{A}(c,t)} R_i \cdot S_i \cdot \text{clip}\left(1 + 0.5 \cdot \text{TitleHint}_i + 0.2 \cdot \text{LongText}_i + 0.3 \cdot \text{MultiPillar}_i,\ 1,\ 2\right) \cdot I_{\text{inf},i}$$

To derive the index, article scores are summed over the inflation-tagged subset of the corpus, divided by the total article count for the country-period, and scaled by 100. This results in an average inflation risk intensity per article, which rises when inflation narrative becomes more frequent, more severe, and more credible, and is insulated from mechanical differences in media volume across countries.

To test the index derived, 46 countries (because of data limitations) from 2015 to 2025 were used to check the statistical relationship between the AIRI and realised inflation.

3.2 Empirical Analysis of AIRI: Stylised Facts on Inflation Risk vs. Inflation Rate Across Countries

This section presents the full set of stylised-fact charts comparing the AIRI inflation-risk indicator with headline CPI inflation across 46 economies, with an interpretation placed directly beneath each chart (or group of charts making the same point).

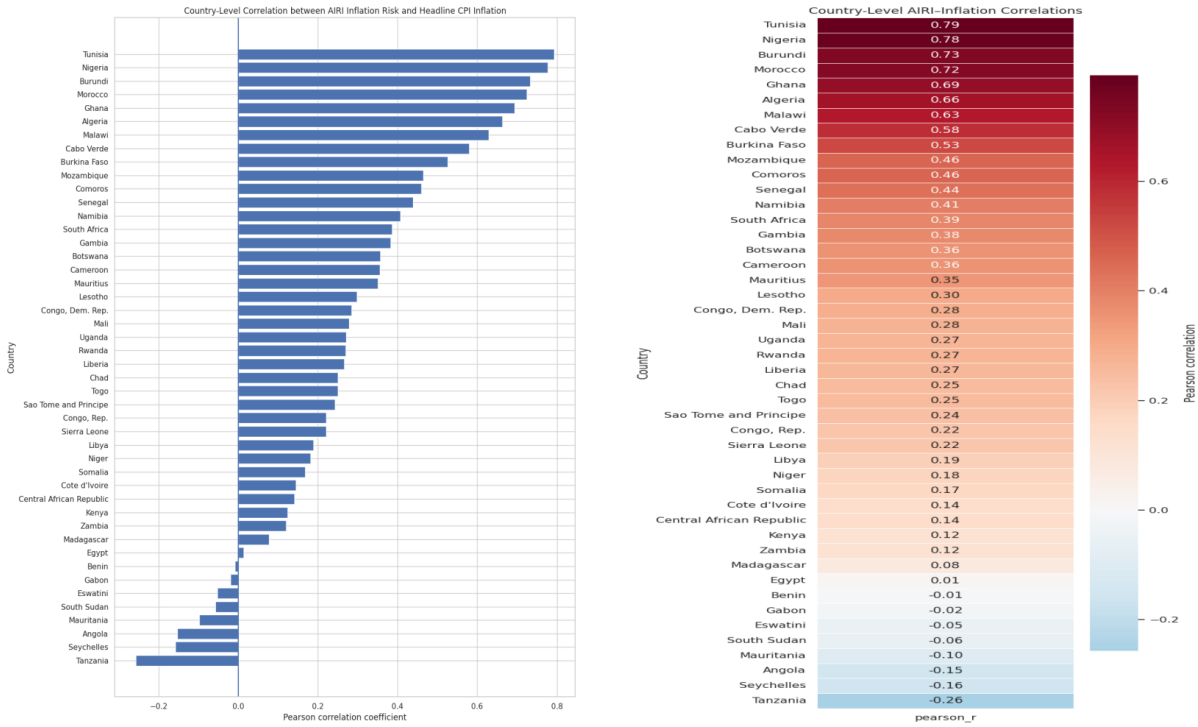


Figure 1. Country-level Pearson correlation between AIRI inflation risk and headline CPI inflation, ranked bar chart (left) and colour-coded heat map (right).

3.2.1 Country-Level Correlation Between AIRI and Headline Inflation

There is considerable cross-country variation. Tunisia (r = 0.793), Nigeria (r = 0.777), Burundi (r = 0.733), Morocco (r = 0.724), Ghana (r = 0.693), Algeria (r = 0.663), Malawi (r = 0.629), Cabo Verde (r = 0.580), and Burkina Faso (r = 0.526) show particularly strong positive relationships between AIRI inflation risk and realised inflation, meaning increases in the raw inflation-risk measure tend to coincide relatively closely with increases in observed CPI inflation in these economies. Across the full panel, the average country-level correlation is 0.281, and the median is 0.268; 38 of 46 countries (82.6%) show a positive relationship, and 32 are statistically significant at the 5% level. Only nine countries clear the "strong" threshold of $r \geq 0.50$, while a handful of countries like Tanzania at r = −0.257, along with Seychelles, Angola, Mauritania, South Sudan, Eswatini, Gabon and Benin — show a negative or essentially flat relationship. In short, AIRI carries real signal about realised inflation in most countries, but its reliability is highly country-specific rather than universal.

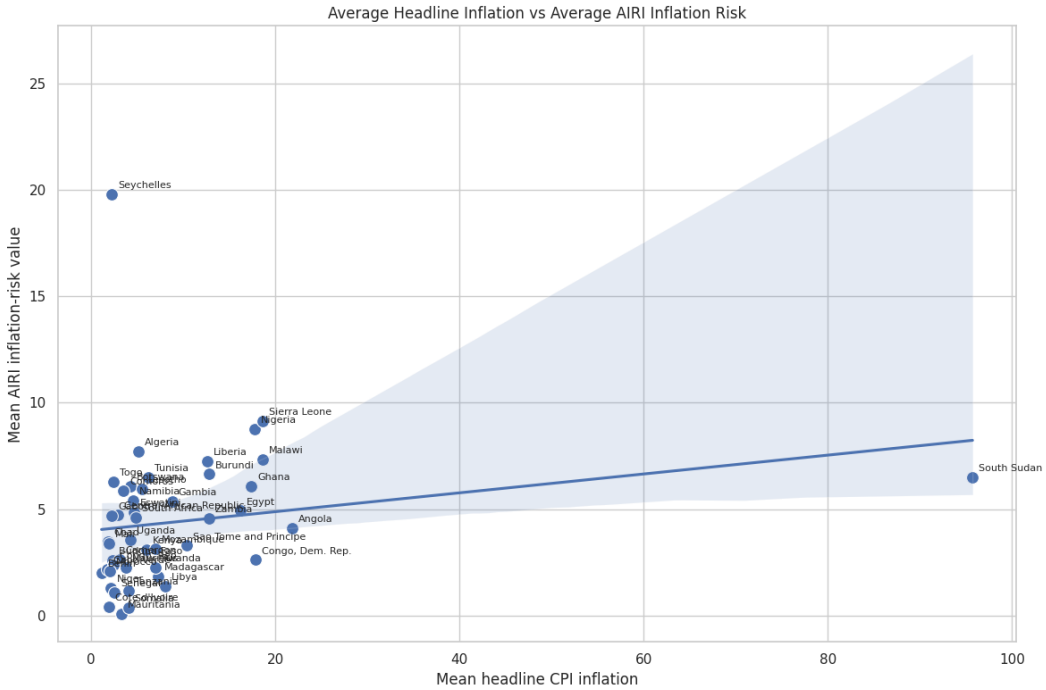


Figure 2. Mean headline CPI inflation versus mean AIRI inflation-risk value, one point per country, with a fitted regression line.

3.2.2 Cross-Country Relationship Between Average Inflation and Average AIRI Risk

At the level of country averages, higher long-run inflation is associated with a higher average AIRI reading, but the fit is loose and is heavily influenced by a few outliers. Seychelles sits far above the line, combining a low mean CPI (around 4%) with an unusually high mean AIRI value (around 20), while South Sudan sits far to the right, with an extreme mean CPI (around 96%) but only a moderate AIRI reading (around 6.5). The other 44 countries cluster tightly in the lower-left corner, with mean CPI below roughly 20% and mean AIRI below roughly 10. This means the overall positive slope is driven disproportionately by a small number of extreme cases rather than reflecting a tight, general relationship between typical inflation levels and typical AIRI readings.

3.2.3 Distribution of the Pooled Panel

Pooling all country-month observations together, both series are heavily right-skewed. The interquartile box for CPI inflation sits close to single digits, but a long tail of outliers extends beyond 500%, driven chiefly by hyperinflation episodes such as South Sudan's. AIRI risk shows the same shape on a more compressed scale: a narrow box roughly between 0 and 6, with outliers reaching above 50. Because both variables spend most of

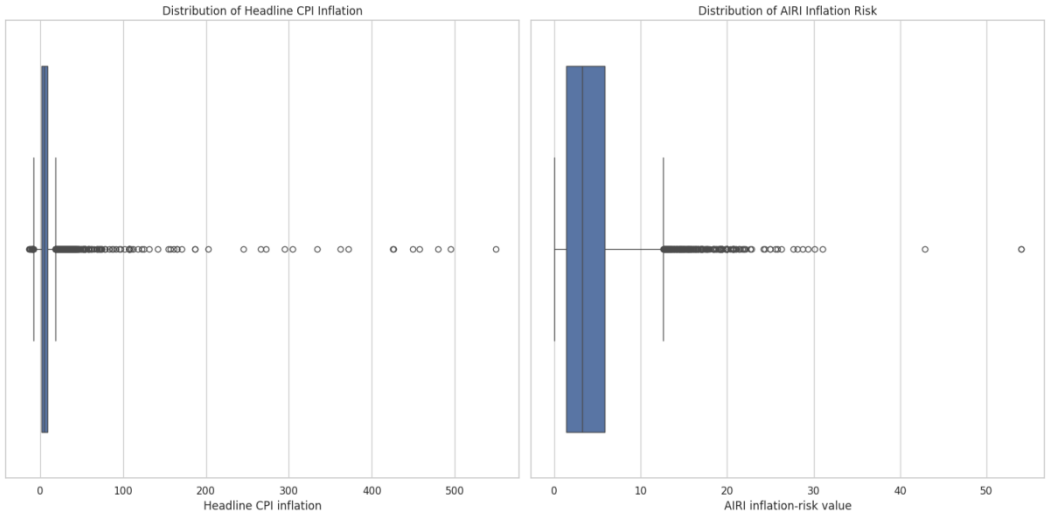


Figure 3. Box plots of headline CPI inflation (left) and AIRI inflation risk (right) pooled across all countries and time periods.

their time in a narrow "normal" range and are dominated in the tails by a small number of crisis episodes, the strength of the AIRI–CPI relationship in any given country depends heavily on how well those tail episodes line up between the two series, which helps explain the wide dispersion of country-level correlations seen in Figure 1.

3.2.4 Country-by-Country Spread of CPI Inflation and AIRI Risk

Breaking the pooled distribution down by country shows that extreme inflation is concentrated in a small number of economies. South Sudan is in a class of its own for CPI inflation, with a median above 50% and outliers beyond 500%, while most other countries have compact boxes below roughly 20–30%; Angola, Congo (Dem. Rep.), Egypt, Ghana, Sierra Leone and Malawi form a second tier of moderately high and volatile inflation. The AIRI distribution tells a related but not identical story: Nigeria, Sierra Leone, Malawi, Comoros, Liberia and Burundi display the widest AIRI boxes, and Seychelles has the single most extreme AIRI outlier (above 50) despite not standing out at all in the CPI chart; a further sign, consistent with Figure 2, that the AIRI measure and realised inflation can diverge sharply for particular countries.

3.2.5 Panel-Wide Trend Over Time

The panel-wide annual averages trace two inflation cycles: a peak around 2016 and a much larger one from 2022 onward, following the global post-pandemic inflation shock,

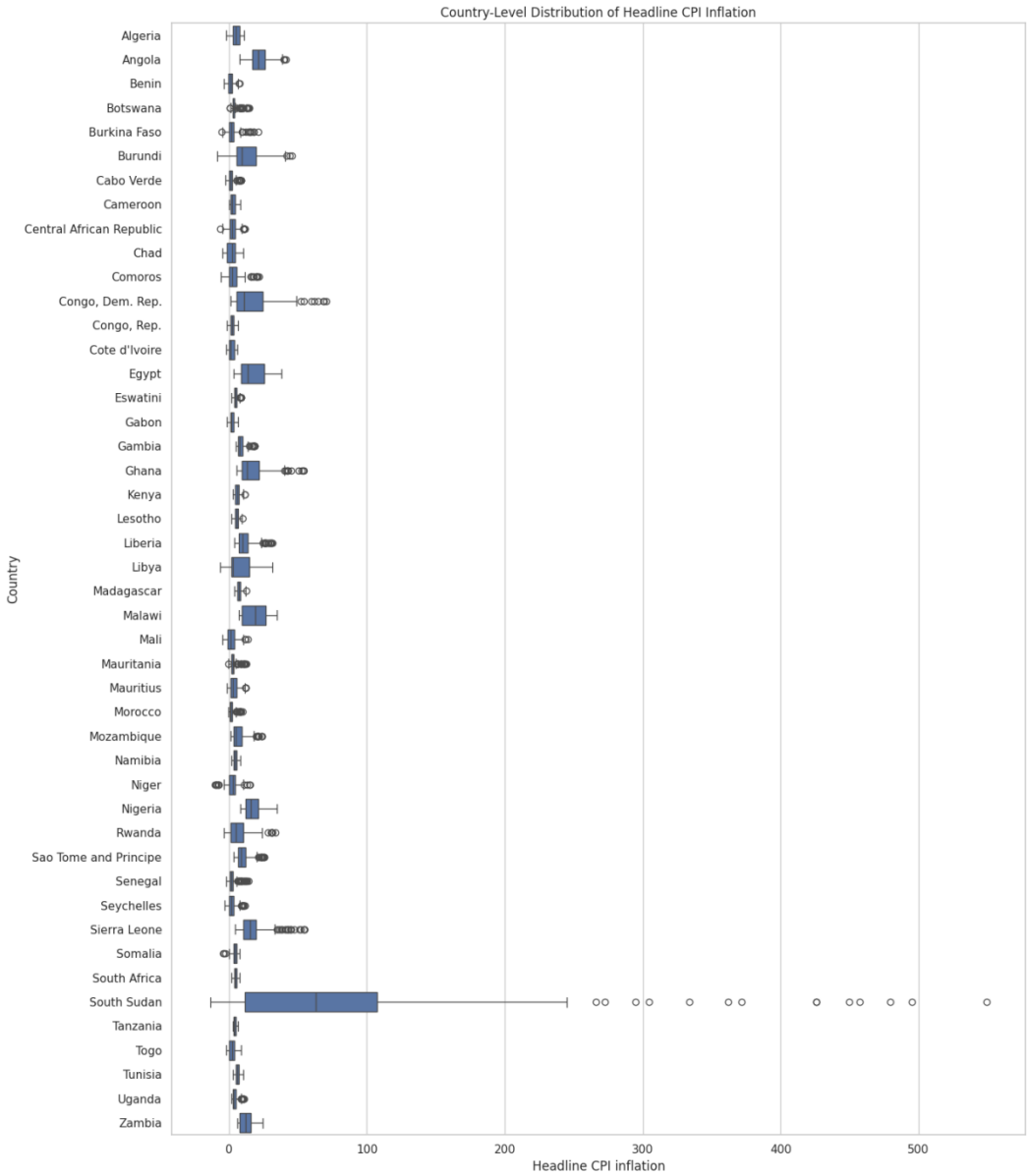


Figure 4a. Country-level distribution of headline CPI inflation.

with a trough in between around 2018–2021. AIRI risk tracks the second episode much more closely than the first. In standardised terms, CPI jumps sharply in 2016 ($z \approx 1.9$) while AIRI barely moves and is even slightly below its own long-run average ($z \approx -0.5$); but in 2022 both series spike together, with AIRI actually showing the larger standardised move ($z \approx 1.9$ versus 0.3 for CPI). After 2022, CPI falls back toward its historical average faster than AIRI, which stays elevated (z of roughly 0.9–1.3) through 2023–2025. This pattern suggests the AIRI measure was particularly responsive to the recent global inflation shock, but is more sluggish than realised inflation both in missing earlier, more localised episodes and in signalling the subsequent disinflation.

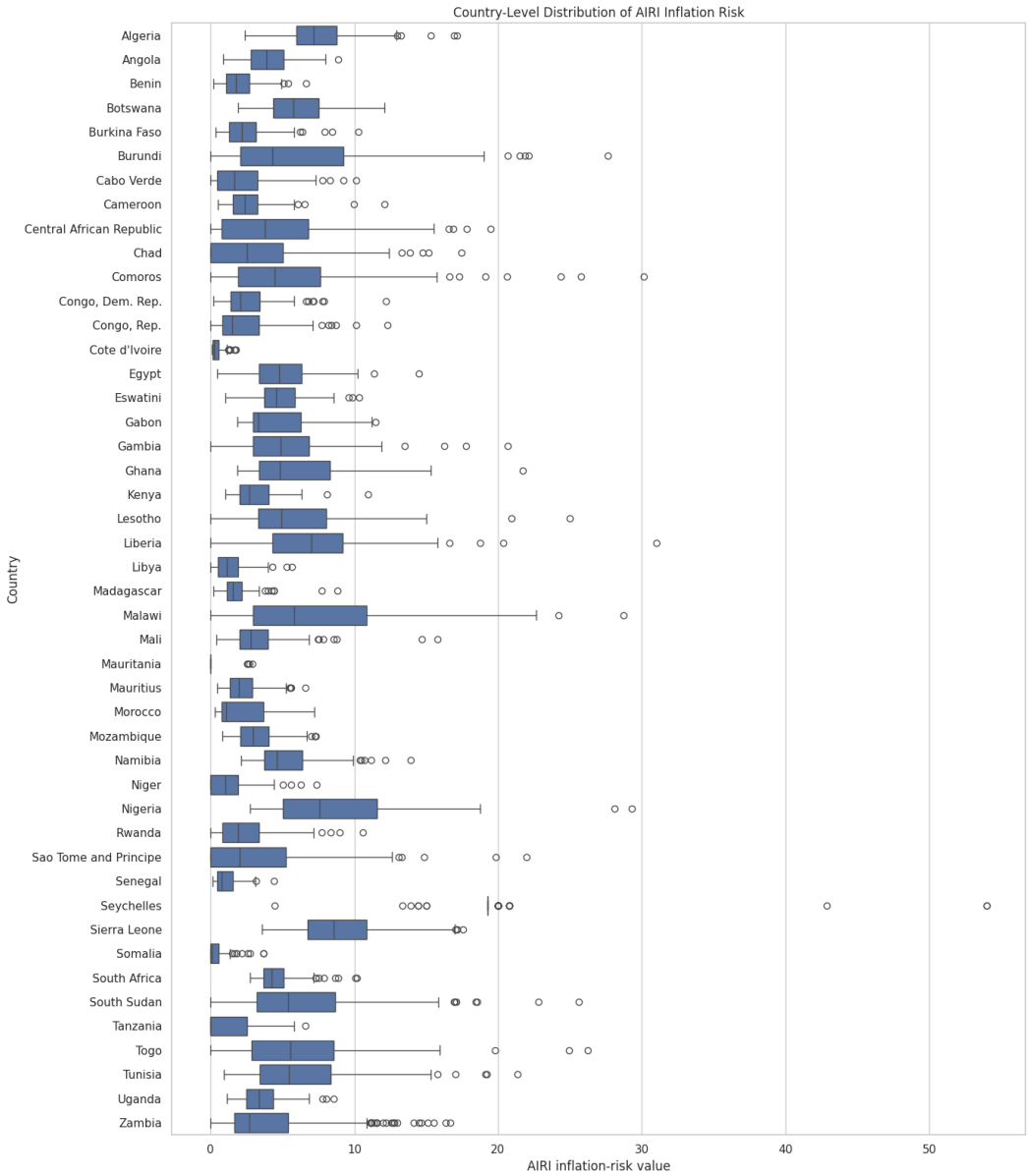


Figure 4b. Country-level distribution of AIRI inflation risk.

3.2.6 Countries Where AIRI Tracks Inflation Well versus Poorly

Contrasting the six strongest-relationship countries (Tunisia, Nigeria, Burundi, Morocco, Ghana, Algeria) with the six weakest (Tanzania, Seychelles, Angola, Mauritania, South Sudan, Eswatini) shows clearly where the aggregate correlation in Figure 1 comes from. In the strong-relationship countries, the two lines rise and fall together and share common turning points, most visibly around the 2022 spike. For example, Tunisia's AIRI jumps to about 15 alongside CPI's jump to about 8.4, and Algeria's two series peak together in 2022 before declining together into 2025. In the weak-relationship countries, the series are effectively decoupled: in Mauritania and Tanzania, AIRI sits near zero for almost the

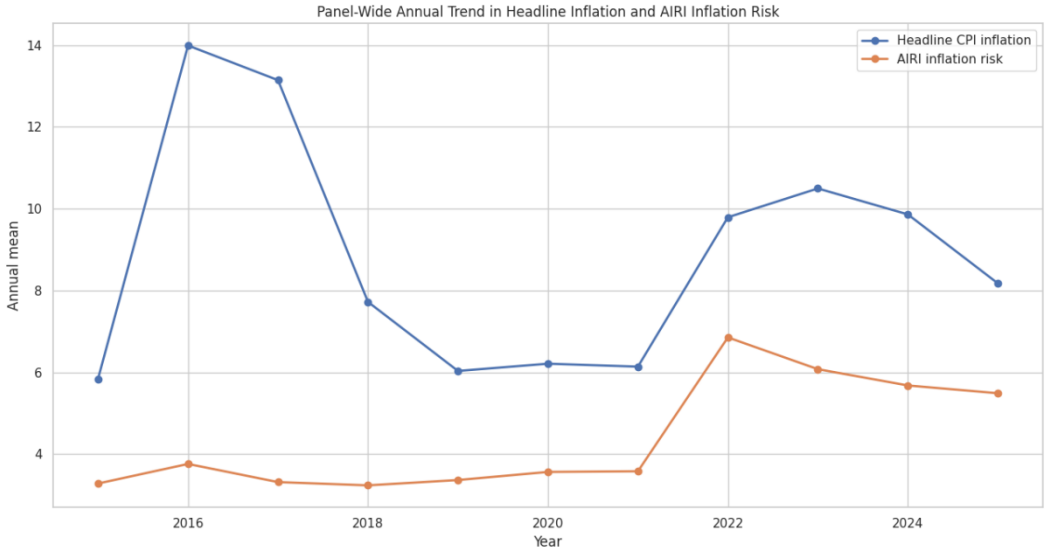


Figure 5a. Annual panel-wide mean of headline CPI inflation and AIRI inflation risk, raw units.

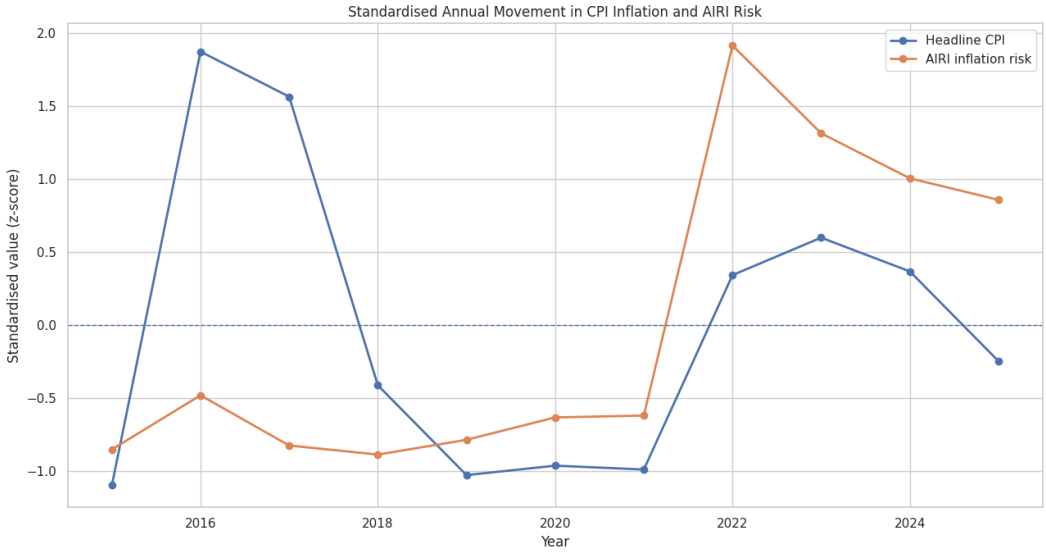


Figure 5b. The same two series expressed as standardised (z-score) annual movements.

whole sample regardless of what CPI is doing; in Seychelles, AIRI stays persistently high (18–20) while CPI oscillates mildly between about −1% and 10%; and in South Sudan, CPI swings wildly — from roughly 350% in 2016 down to near zero and back above 100% by 2025 — while AIRI barely registers any movement. This suggests AIRI behaves more like a slow-moving structural risk signal in some countries, one that can become almost uninformative, or even move independently of, realised inflation when that inflation is driven by idiosyncratic shocks, currency collapse, or conflict.

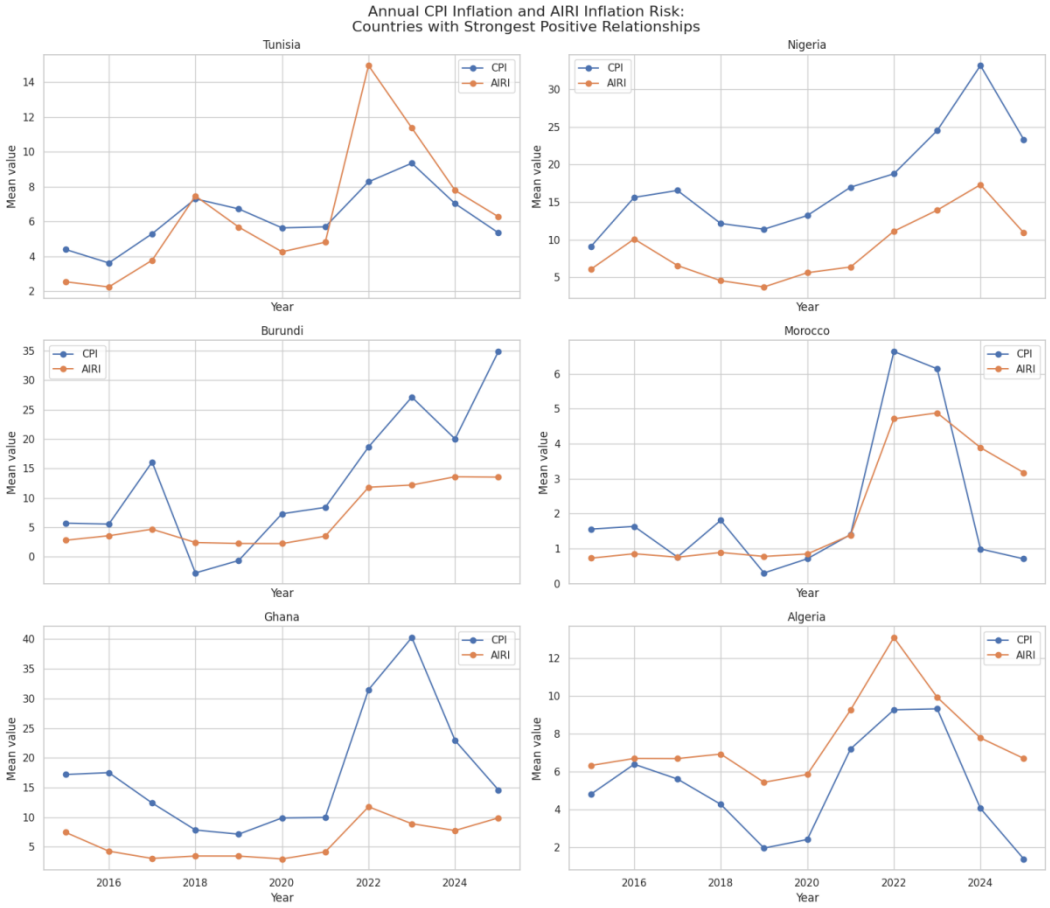


Figure 6a. Annual CPI inflation and AIRI risk for the six countries with the strongest AIRI–CPI relationship.

3.2.7 Shape of the AIRI–Inflation Relationship in the Top Countries

Looking at the underlying scatter of monthly observations rather than the single correlation coefficient in Figure 1 shows that even the "top 12" countries differ in how reliable their relationship is. Tunisia, Nigeria and Morocco show a fairly linear, tightly banded relationship with narrow confidence bands around the fitted line, indicating a genuinely proportional co-movement. Others further down the ranking, such as Cabo Verde, Comoros and Senegal, show noisier clouds of points, often with many observations clustered near zero on both axes and only a handful of higher-inflation episodes pulling the regression line upward. This confirms that a "positive and significant" country correlation can reflect either a consistently proportional relationship or one driven mainly by a few extreme episodes within an otherwise flat pattern.



Figure 6b. Annual CPI inflation and AIRI risk for the six countries with the weakest AIRI–CPI relationship.

3.2.8 Volatility Comparison

Ranking countries by the standard deviation of CPI inflation against that of AIRI risk shows that CPI inflation is, with virtually no exceptions, far more volatile than AIRI. The gap is most extreme for South Sudan, where CPI volatility (≈120) is more than twenty times AIRI volatility (≈5), but the same pattern holds for Congo (Dem. Rep.), Sierra Leone, Libya, Egypt, Ghana and Burundi as well. Because AIRI is a structurally smoother, lower-variance series than realised CPI inflation, it is mechanically unable to track the sharpest inflation swings even in countries where the two series are correlated on average, which is part of why even the strongest country correlations in Figure 1 still leave much of the variation in CPI unexplained.

3.2.9 Macroeconomic Correlation Matrix

At the pooled macroeconomic level, money growth, currency depreciation, CPI inflation, AIRI risk and real GDP growth are all only weakly correlated with one another since

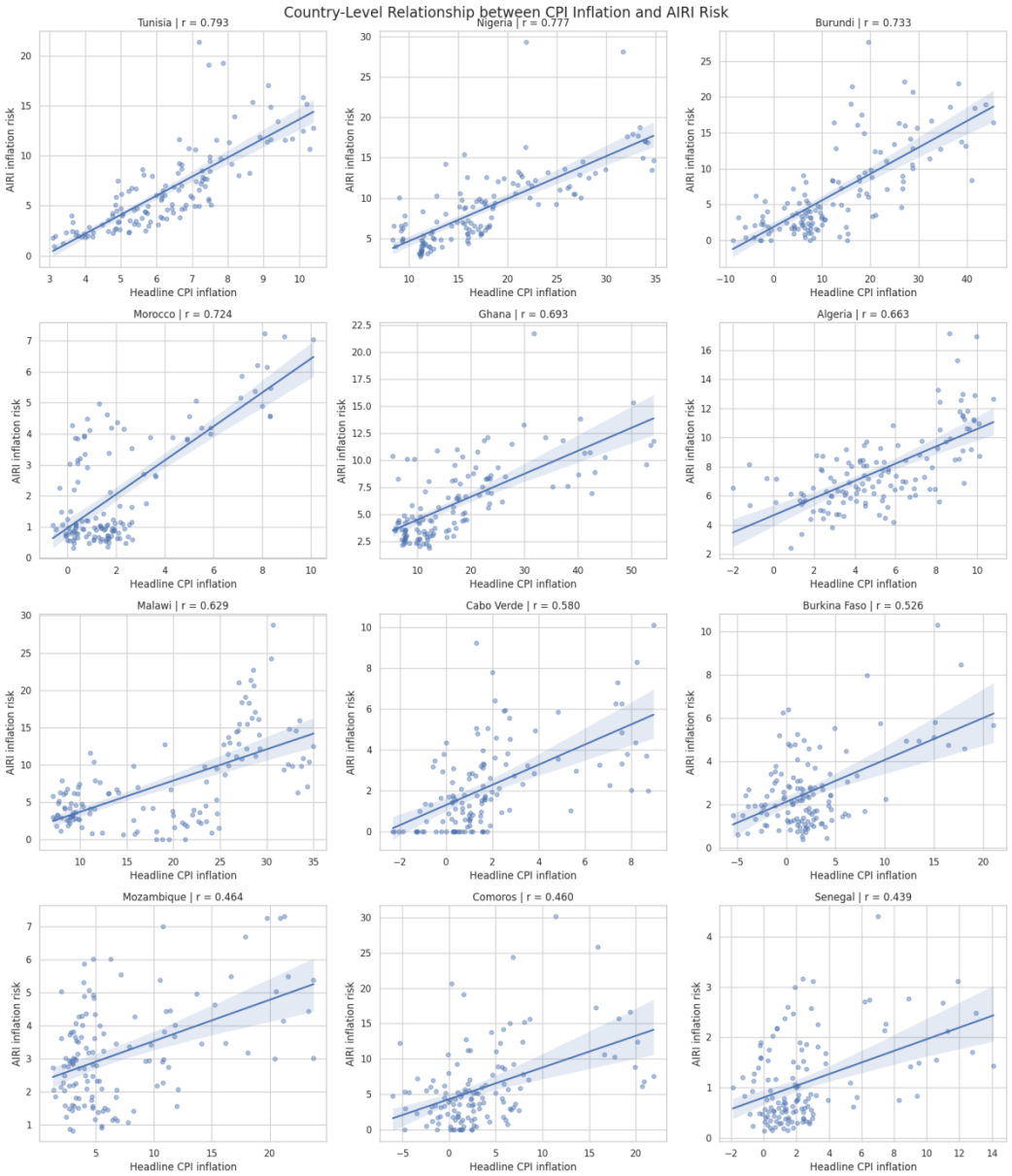


Figure 7. Country-level scatter plots of AIRI risk against headline CPI inflation, with fitted regression line, for the twelve countries with the strongest correlations.

Table 1: Pooled Pairwise Correlation Matrix

	m	**Δe**	π	**AIRI**	**g**
m	1.000	0.206	0.082	−0.013	−0.082
Δe	0.206	1.000	0.171	0.059	−0.067
π	0.082	0.171	1.000	0.129	−0.168
AIRI	−0.013	0.059	0.129	1.000	−0.048
g	−0.082	−0.067	−0.168	−0.048	1.000

Note: Pooled pairwise correlations between money growth (m), currency depreciation (Δe), CPI inflation (π), AIRI risk (AIRI) and real GDP growth (g).

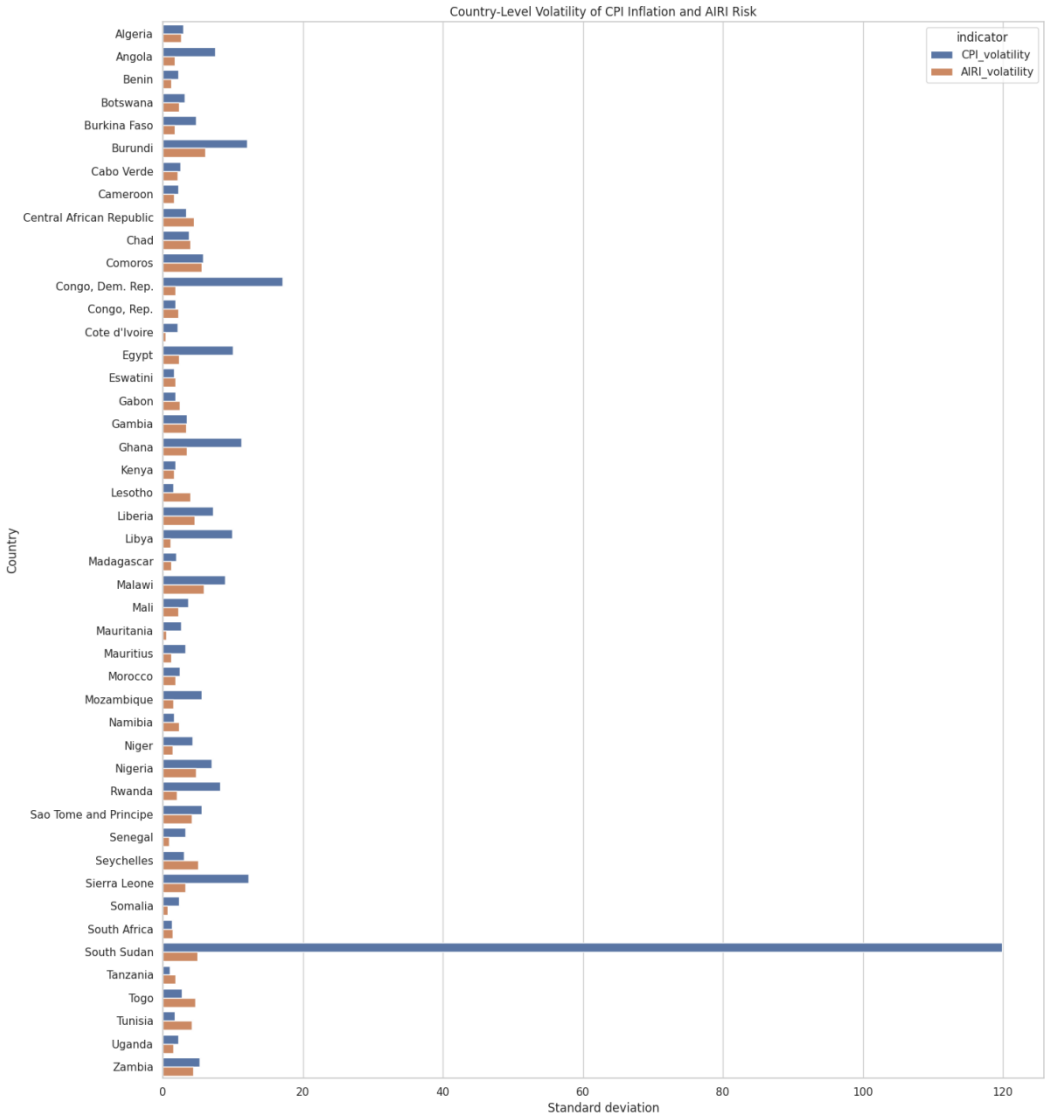


Figure 8. Country-level standard deviation (volatility) of headline CPI inflation versus AIRI inflation risk.

no off-diagonal coefficient exceeds 0.21 in absolute value. Money growth and currency depreciation have the strongest pairwise link (0.206), consistent with standard monetary and exchange-rate channels of inflation, and inflation is negatively correlated with growth (-0.168), consistent with a growth–inflation trade-off. Notably, the correlation between the AIRI inflation risk pillar and headline CPI inflation at this pooled, cross-country level is only 0.129, which is much weaker than the 0.281 average correlation reported country-by-country in Figure 1.

3.2.10 Time-Varying (Rolling) Correlation

The rolling correlation shows that even for the four best-performing countries in the full-sample ranking, the AIRI–CPI relationship is far from stable through time. All

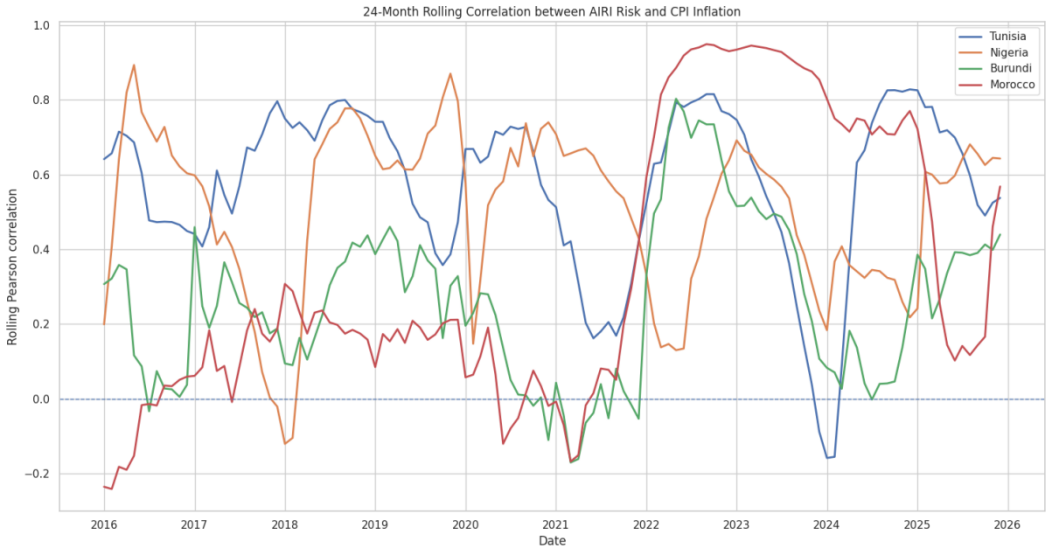


Figure 9. 24-month rolling Pearson correlation between AIRI risk and CPI inflation for Tunisia, Nigeria, Burundi and Morocco.

four lines fluctuate substantially, occasionally swinging from above 0.8 to near zero or negative within a couple of years — Tunisia's rolling correlation, for instance, falls to about −0.16 in 2024 before recovering above 0.8 by 2025. The clearest common feature is a broad convergence toward high correlation (roughly 0.7–0.95) during the 2022–2023 global inflation shock, when all four lines rise together; outside that window the countries behave much more idiosyncratically, at times differing from one another by 0.5 or more at the same point in time. This confirms that the AIRI–CPI relationship is regime-dependent: it strengthens markedly during large, common inflation shocks and weakens or becomes unstable in quieter periods.

3.2.11 Where AIRI Over- or Under-States Realised Inflation

Ranking countries by the gap between mean AIRI risk and mean CPI inflation highlights differences in level, separately from the correlation results above. Seychelles is the only country where AIRI substantially overstates inflation relative to CPI, by roughly 18 points on average. At the other extreme, South Sudan, Angola, Congo (Dem. Rep.), Egypt, Malawi, Ghana and Sierra Leone all show AIRI running far below realised inflation, which is most dramatic in South Sudan, where the gap is about −90 points, reflecting hyperinflation episodes that the AIRI measure comes nowhere close to matching in scale. The majority of countries in the middle of the chart cluster close to zero, meaning AIRI and CPI are of a broadly similar order of magnitude in most "normal-inflation" economies, with the largest gaps concentrated in the same handful of extreme or crisis-affected countries that also stand out in Figures 2, 4 and 8.

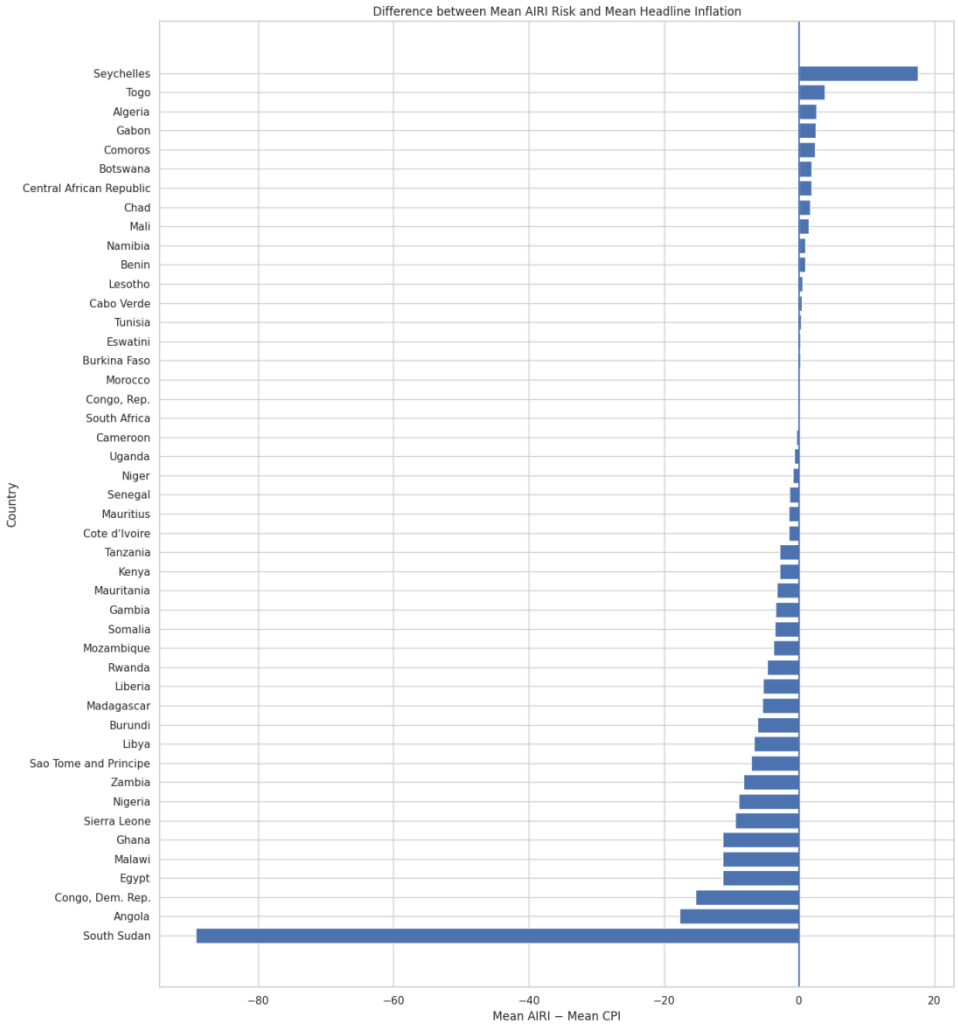


Figure 10. Difference between each country's mean AIRI risk and its mean headline CPI inflation, ranked.

Table 2: Country-Level Summary Statistics

Metric	**Value**
Number of countries	46
Average country-level Pearson correlation	0.281
Median country-level Pearson correlation	0.268
Countries with positive AIRI–CPI correlations	38 (82.6%)
Countries with negative correlations	8
Countries with statistically significant correlations (5%)	32
Countries with strong positive correlations ($r \geq 0.50$)	9
Strongest positive relationship	Tunisia ($r = 0.793$)
Weakest relationship	Tanzania ($r = -0.257$)

3.2.12 Country-Level Summary

Taken together, these headline statistics support an overall reading of the stylised facts above: AIRI carries genuine, statistically detectable signal about realised inflation in most economies (a positive relationship in 82.6% of countries, significant in 32 of 46), but that signal varies enormously in strength, is diluted once countries are pooled together, is structurally less volatile than realised inflation, and can weaken, become unstable over time, or break down entirely in a minority of countries with unusual inflation dynamics such as hyperinflation, currency crises, or conflict.

3.3 Panel VAR Methodology

To investigate whether the AIRI predicts realised inflation when other factors have been accounted for, we estimate a panel vector autoregressive model for all 46 countries in Africa. The Panel VAR captures the dynamic relationship between variables, considering all variables as endogenous. The estimation allows for the computation of the direction of the statistical relation through Granger causality. The response of realised inflation to changes in the AIRI will be computed by the Impulse Response Function (IRF).

3.3.1 Model Specification

Following Love and Zicchino (2006) and Canova and Ciccarelli (2013), the reduced-form Panel VAR of order p is specified as:

$$Y_{c,t} = A_0(c) + \sum_{k=1}^{p} A_k Y_{c,t-k} + u_{c,t}$$

where $c = 1, \ldots, N$ indexes countries ($N = 46$) and $t = 1, \ldots, T$ indexes monthly time periods (January 2015 to December 2025). $Y_{c,t}$ is the 5×1 vector of endogenous variables:

$$Y_{c,t} = \left[\text{AIRI}\pi_{c,t}, \Delta e_{c,t}, m_{c,t}, g_{c,t}, \pi_{c,t}\right]^{\top}$$

where $\text{AIRI}\pi_{c,t}$ is the AIRI inflation pillar score for country c at time t, scaled to the interval $[0, 200]$ and normalised by total article volume, and $\pi_{c,t}$ is year-on-year headline CPI inflation. We identify other variables that have been tested in the literature to predict inflation, in order to account for factors that can influence realised inflation: $\Delta e_{c,t}$ is the month-on-month percentage change in the nominal exchange rate, local currency per USD (Razafimahefa 2012); $m_{c,t}$ is money supply (M2) growth (Okedigba et al. 2024;

Table 3: Variable Definitions and Data Sources

Variable	Definition	Source
AIRI$\pi(c,t)$	AIRI inflation pillar score, country c, month t. Normalised by total article volume; scale 0–200	Derived index
$\pi(c,t)$	Year-on-year headline CPI inflation (%)	African Development Bank (AfDB)
$\Delta e(c,t)$	Month-on-month change in nominal exchange rate, local currency/USD (%)	African Development Bank (AfDB)
$g(c,t)$	Real GDP growth (%)	African Development Bank (AfDB)
$m(c,t)$	Money supply (M2) growth (%)	African Development Bank (AfDB)

Alimi 2012); and $g_{c,t}$ is real GDP growth (Barnichon and Peiris 2007). All five variables are treated as endogenous, so each can respond to shocks in any of the others. $A_0(c)$ is a 5×1 vector of country-specific intercepts, capturing the fact that countries differ in their average level of risk narrative and inflation regardless of dynamics. A_k for $k = 1, \ldots, p$ are 5×5 matrices of lag coefficients, and $u_{c,t}$ is the 5×1 vector of reduced-form disturbances, with $E\left[u_{c,t}\right] = 0$, $E\left[u_{c,t}u'_{c,t}\right] = \Sigma$, and $E\left[u_{c,t}u'_{c,s}\right] = 0$ for $t \neq s$.

Slope homogeneity is tested with the Pesaran and Yamagata (2008) test, a standardised version of Swamy's (1970) statistic. For an equation with k slope coefficients:

$$\widetilde{\Delta} = \frac{\sqrt{N}\left(N^{-1}\widetilde{S} - k\right)}{\sqrt{2k}}, \qquad \widetilde{S} = \sum_{c}\left(\hat{\beta}_c - \tilde{\beta}_{WFE}\right)'\left(\frac{X'_c M X_c}{\tilde{\sigma}_c^2}\right)\left(\hat{\beta}_c - \tilde{\beta}_{WFE}\right)$$

where $\hat{\beta}_c$ is the vector of slope coefficients of country c in the equation being tested, $\tilde{\beta}_{\mathrm{WFE}}$ is the weighted fixed-effects estimate, $X'_c M X_c$ is the country's demeaned moment matrix, and $\tilde{\sigma}_c^2$ is its error variance estimated from the pooled fixed-effects residuals. Under the null of slope homogeneity $\widetilde{\Delta}$ is asymptotically standard normal.

This was done to determine whether to estimate the panel as pooled or Mean Group. Mean Group places no restriction on A_k across countries and remains consistent whether or not slope homogeneity holds, whereas the pooled fixed-effects estimator is consistent only if homogeneity holds.

3.3.2 Pre-Estimation Diagnostics

Before estimation, each variable is tested for stationarity using the Im, Pesaran, and Shin (2003) panel unit root test, which allows the autoregressive coefficient to vary across countries (Kelikume 2018; Phiri 2016). Where a series is non-stationary in levels, first differences are used and long-run relationships are assessed with the Westerlund (2007)

panel cointegration test.

Cross-sectional dependence is tested using the Pesaran (2004) CD test, applied to each variable's raw univariate autoregression to decide whether the series exhibit cross-sectional dependence and to select the appropriate unit root test to adopt.

3.3.3 Granger Causality

To test whether AIRI statistically precedes inflation, a Granger causality test is carried out. The null hypothesis of non-causality from AIRIπ to π is:

$$H_0 : A_k \left[\text{AIRI}\pi \to \pi\right] = 0 \quad \text{for all } k = 1, \ldots, p$$

Given the cross-country heterogeneity already established, the Dumitrescu and Hurlin (2012) test is used rather than a pooled panel Granger test, since a pooled test would impose a common causal coefficient across all 46 countries and understate the relationship in countries where it is in fact strong. The test computes a country-specific Wald statistic for each country and averages them:W^c

$$\bar{W} = N^{-1} \sum_c W^c$$

Its standardised form is asymptotically standard normal under the null of homogeneous non-causality as $N, T \to \infty$. Rejecting the null establishes that AIRI carries predictive information for inflation beyond inflation's own history—the paper's central empirical claim, and one that does not depend on any of the identifying assumptions in Section 3.3.4.

3.3.4 Impulse Response Function (IRF) Analysis

Granger causality establishes whether AIRI predicts inflation, but not the size or duration of that relationship. Impulse Response Functions are used to trace the path of inflation following a one-time shock to AIRI (Canova and Ciccarelli 2013).

Because the reduced-form residuals $u_{c,t}$ are correlated across the five equations, a shock cannot be isolated to one variable without first imposing a causal ordering. A Cholesky decomposition of the error covariance matrix Σ is used for this purpose (Love and Zicchino 2006), under the ordering:

$$Y_{c,t} = \left[\text{AIRI}\pi_{c,t}, \pi_{c,t}, \Delta e_{c,t}, m_{c,t}, g_{c,t}\right]^\top$$

To evaluate the response of inflation to a change in AIRI, AIRI is ordered first. We also confirm that the ordering of the other four variables does not affect the result, by changing the ordering of the other four variables.

Further, as a sensitivity check on the identifying assumption itself, we place AIRI last, so that it cannot affect any other variable within the month. Under this ordering the impact response is zero by construction. The results are reported in Section 4.4 and Appendix Table A1.

$$Y_{c,t}^{alt} = \left[g_{c,t}, m_{c,t}, \Delta e_{c,t}, \pi_{c,t}, \text{AIRI}\pi_{c,t}\right]^{\top}$$

3.4 Forecast Evaluation

This section seeks to establish the forecasting value of the AIRI inflation pillar. This is achieved through an out-of-sample forecast evaluation to test whether including the AIRI inflation pillar in a benchmark model improves forecast accuracy relative to benchmark models.

The forecast evaluation is structured around a nested model comparison. Three panel regression models are specified, each adding one additional layer of information. This enables us to trace any improvement in forecast accuracy to the specific change implemented in the new model, moving from the AR benchmark to a model that includes the AIRI inflation pillar.

The AR benchmark follows the proposition of Stock and Watson (1999) and Atkeson and Ohanian (2001) that the naïve method of forecasting inflation from its own past values is difficult to outperform, and should serve as the standard that any meaningful model must supersede. The model is specified as:

$$\pi_{c,t+h} = \alpha_c + \sum_{i=1}^{p-1} \gamma_i \pi_{c,t-i} + \varepsilon_{c,t+h}$$

where $\pi_{c,t+h}$ is inflation in country c at horizon h months ahead, α_c denotes country fixed effects, p is the optimal lag length selected by the Bayesian Information Criterion, and $\varepsilon_{c,t+h}$ is the error term.

The macro benchmark extends Model 0 by incorporating a standard set of macroeconomic control variables—the nominal exchange rate, money supply, and GDP growth—known to influence inflation dynamics in African economies (Nguyen et al. 2017; Alper, Hobdari, and Uppal 2016):

$$\pi_{c,t+h} = \alpha_c + \sum_{i=1}^{p-1} \gamma_i \pi_{c,t-i} + \sum_{j=1}^{J} \delta_j X_{j,c,t} + \varepsilon_{c,t+h}$$

where $X_{j,c,t}$ denotes the j-th macroeconomic control variable observed at time t and δ_j is its coefficient.

The AIRI model further augments Model 1 with the AIRI inflation pillar:

$$\pi_{c,t+h} = \alpha_c + \sum_{i=1}^{p-1} \gamma_i \pi_{c,t-i} + \sum_{j=1}^{J} \delta_j X_{j,c,t} + \beta \cdot \mathrm{AIRI}_{c,t} + \varepsilon_{c,t+h}$$

where $\mathrm{AIRI}_{c,t}$ is the AIRI inflation pillar observed at time t and β is the key coefficient capturing the incremental predictive contribution of AIRI for future inflation after conditioning on inflation persistence and the standard macroeconomic information set. A positive and statistically significant β indicates that the AIRI inflation pillar contains genuine predictive information beyond what conventional macroeconomic variables already provide.

All three models are estimated separately at forecast horizons $h \in \{1, 3, 6, 12\}$ months by pooled OLS using a recursive expanding window, in which the estimation sample expands forward through time at each forecast origin—a design that mirrors the information set available to a real-time forecaster (Stock and Watson 1999).

Forecast accuracy is evaluated using the Root Mean Squared Error (RMSE).

$$\mathrm{RMSE}_m = \sqrt{\frac{1}{N \cdot T_{test}} \sum_{c=1}^{N} \sum_{\tau \in \mathcal{T}} \left(\pi_{c,\tau+h} - \hat{\pi}^m_{c,\tau+h} \right)^2}$$

Also, the Clark–West MSFE-adjusted statistic was employed. This approach accounts for the sampling noise generated by the nested model. Because Model 2 nests Model 1, the standard Diebold–Mariano test is biased against detecting genuine forecast improvements in nested comparisons (Clark and West 2007). The Clark–West MSFE-adjusted statistic is therefore used as the primary test:

$$\hat{d}_t = \left(\pi_{c,t+h} - \hat{\pi}^1_{c,t+h} \right)^2 - \left[\left(\pi_{c,t+h} - \hat{\pi}^2_{c,t+h} \right)^2 - \left(\hat{\pi}^1_{c,t+h} - \hat{\pi}^2_{c,t+h} \right)^2 \right]$$

$$CW = \frac{\bar{d}}{\hat{\sigma}_d / \sqrt{T_{test}}}$$

The null is rejected if $CW > 1.645$ at the 5 per cent significance level.

Table 4: Pesaran Cross-Sectional Dependence (CD) Test Results

Variable	CD statistic	p-value	Dependence present
Inflation (π)	82.3	0.000	Yes
AIRI Pillar	84.1	0.000	Yes
GDP Growth	120.0	0.000	Yes
Money Growth	21.1	0.000	Yes
FX Growth	78.4	0.000	Yes

4 Results and Discussion

4.1 Pre-Estimation Diagnostics

4.1.1 Cross-Sectional Dependence

The choice of unit root test depends on whether the panel exhibits cross-sectional dependence, so the Pesaran (2004) CD test is conducted first. The test examines the average pairwise correlation of residuals across countries. This step is necessary because the Im, Pesaran and Shin (2003) test assumes that each country's process is independent of the others, whereas neighbouring African economies share exposure to common commodity, food and global financial shocks and are likely to move together when such shocks occur. Under the null hypothesis of no cross-sectional dependence ($\rho_{ij} = 0$ for all $i \neq j$), the CD statistic is asymptotically standard normal. The CD test is carried out at the 5% two-sided nominal significance level; the null is rejected if the |CD| statistic is greater than 1.96.

The panel exhibits significant cross-sectional dependence (Pesaran 2004). Ignoring such dependence distorts the size of panel unit root tests, which over-reject the unit-root null when countries are correlated (O'Connell 1998). We therefore supplement the standard IPS test (Im, Pesaran, and Shin 2003) with the cross-sectionally augmented IPS test (CIPS; Pesaran 2007).

4.1.2 Panel Unit Root Tests

CIPS augments each country's Dickey–Fuller regression with the cross-sectional averages of the dependent variable and its lagged differences, so that the common factor driving co-movement is partialled out before stationarity is assessed. The computed statistic is compared with the critical values in Pesaran (2007), and the series is stationary if the statistic is more negative than the critical value. For a panel of 46 countries and 132 monthly observations the relevant critical values are approximately -2.11 at the 5 per

Table 5: IPS and CIPS Panel Unit Root Test Results

Variable	IPS statistic	p-value	Rejects unit root?	Conclusion
Inflation (π)	-10.8	0.000	Yes	Stationary
AIRI pillar	-12.4	0.000	Yes	Stationary
GDP growth	-13.5	0.000	Yes	Stationary
Money growth	-48.2	0.000	Yes	Stationary
FX growth	-47.6	0.000	Yes	Stationary

Note: N = 46 countries, monthly data. The statistic is the standardised $W_{\bar{t}}$ of Im, Pesaran and Shin (2003), asymptotically N(0,1) under the null. Country-level ADF regressions include an intercept, with lag length chosen by AIC for each country. Null hypothesis: all countries have a unit root. Alternative: at least some countries are stationary. The null is rejected for large negative values.

cent level and -2.23 at the 1 per cent level. The IPS test was also done for robustness; in this case, the p-value is the indicator of stationarity.

Tables 4 and 5 show that both tests agree, indicating that the existence of cross-sectional dependence between countries does not affect the stationarity of our series.

Both tests agree that all five series are stationary in levels, so the panel VAR is estimated in levels and no differencing is required. Because stationarity is not rejected for any series, panel cointegration testing is not applicable and the Westerlund (2007) procedure described in Section 3.3.2 is not carried out.

Two qualifications are recorded. First, the CIPS statistic for inflation, at -2.29, clears the 1 per cent critical value by a narrow margin, which is considerably less comfortable than the margins for the other four series. Headline inflation in this panel is therefore stationary but highly persistent, a finding reflected directly in the autoregressive coefficients of Section 4.2 and relevant to the interpretation of the impulse responses. Second, stationarity here is a statement about the panel as a whole. Individual countries with extreme inflation histories, most obviously South Sudan, may behave differently, and the Mean Group estimator accommodates this by estimating country by country rather than imposing a common process.

4.2 Panel VAR Results

4.2.1 Lag Selection

A common lag order is selected across countries. Separate lag selection was conducted for each country under the Akaike Information Criterion (AIC), the Hannan–Quinn Criterion (HQ) and the Schwarz Criterion (SC/BIC), and the modal recommendation under each was recorded.

Table 6: Lag Selection Criteria (AIC, HQ, SC/BIC)

Criterion	Modal lag
AIC	6 (24/46, 52%)
HQ	3 (36/46, 78%)
SC / BIC	**2 (38/46, 83%)**

Table 7: Slope Homogeneity Tests (Pesaran–Yamagata)

Equation / sample	Statistic	Adjusted statistic	p-value	Reject null?
Inflation (46 countries)	**3.96**	**4.15**	< 0.001	Yes
AIRI (46 countries)	12.66	13.29	< 0.001	Yes
FX growth (46 countries)	5.77	6.05	< 0.001	Yes
Money growth (46 countries)	50.33	52.83	< 0.001	Yes
GDP growth (46 countries)	25.01	26.25	< 0.001	Yes
Inflation, excl. South Sudan (45)	14.03	14.73	< 0.001	Yes
Inflation, Swamy version (46)	22.01	23.10	< 0.001	Yes

The Schwarz criterion is adopted, and a lag order of two is imposed throughout. Two considerations support this. SC/BIC shows the strongest and most consistent agreement across countries, recommending lag 2 for 38 of the 46 countries, or 83 per cent, whereas AIC is modal for only a bare majority. It is also the most parsimonious of the three, which matters materially here: with approximately 130 observations per country and a five-variable system estimated separately for each country, each additional lag consumes 25 coefficients per country. At the AIC recommendation of six lags the country-level systems would require roughly 150 parameters, leaving degrees of freedom too thin to support the country-specific estimates on which both the Mean Group average and the Dumitrescu–Hurlin test depend. Results under the Hannan–Quinn selection of three lags are reported in the Appendix and are qualitatively unchanged.

4.2.2 Poolability Test

Slope homogeneity is tested with the Pesaran and Yamagata (2008) test to check for the appropriate approach for the panel VAR analysis. The test answers the question: can the countries be represented with the same coefficient? Specifically, the null hypothesis states that one coefficient can describe the panel. The result shows that homogeneity is rejected in every equation (Table 7). This suggests that the best approach to the panel is the Mean Group estimator.

4.2.3 Mean Group Panel VAR Results

The Mean Group estimator is adopted because the panel is heterogeneous in ways that a pooled specification cannot accommodate. Pesaran and Smith (1995) show that imposing a common slope on dynamic processes that genuinely differ across units yields inconsistent estimates, and that the inconsistency does not vanish as the cross-section grows: the misspecification is in the restriction itself, not in the sample size. The Mean Group procedure instead estimates the panel VAR separately for each of the 46 countries, allowing every coefficient to differ, and reports the cross-country average with standard errors derived from the dispersion of those country-level estimates.

The heterogeneity of this panel is pronounced on every dimension relevant to the model. Average headline inflation ranges from 1.15 per cent in Benin to 95.71 per cent in South Sudan, with South Africa at 4.8 per cent and Angola at 21.8 per cent occupying the intermediate ground; a single autoregressive coefficient cannot plausibly describe price dynamics across that range. Monetary and exchange rate arrangements differ correspondingly, from the hard pegs of the two CFA franc zones to the inflation-targeting regime of South Africa and the managed floats and periodic devaluations elsewhere, so the transmission of monetary and exchange rate shocks to prices is structurally different across the panel. Nguyen et al. (2017) document that country-specific characteristics including oil and food import dependence, exposure to weather shocks and the weight of agriculture in the consumption basket determine how inflationary shocks propagate within African economies. Media sectors, which form the base of the AIRI, vary comparably in scale, ownership structure and editorial independence, so the mapping from underlying inflation risk to observed news intensity is itself country-specific. The Mean Group estimator is the natural response to heterogeneity of this kind, and Section 4.2.2 reports the formal poolability test on which the choice rests.

The results show that AIRI is a leading indicator of inflation at lag one, even after controlling for inflation's persistence and the conventional macroeconomic determinants. A one unit increase in AIRI is associated with a 0.08 percentage point increase in next-period inflation at the 5 per cent level. The second lag is insignificant, indicating that the informational value of the index is concentrated in the immediately preceding month, which is the expected signature of a real-time signal. Because inflation in this panel is highly persistent, the cumulative effect is considerably larger than the impact coefficient, on the order of one percentage point per unit of the index. Notably, money growth is insignificant and exchange rate depreciation only marginally significant at the monthly frequency, so AIRI carries predictive information at a horizon where the conventional indicators are largely silent, and does so without the reporting lag those indicators entail. The finding aligns with the growing literature showing that news-based information

Table 8: Mean Group Panel VAR Results — Inflation Equation

Predictor	Estimate	p-value
Inflation (lag 1)	1.160	0.000
Inflation (lag 2)	−0.231	0.000
Intercept	0.213	0.038
AIRI (lag 1)	**0.0801**	**0.040**
GDP Growth (lag 1)	−0.117	0.048
Exchange Rate Growth (lag 1)	0.0357	0.058
GDP Growth (lag 2)	0.0889	0.086
Exchange Rate Growth (lag 2)	0.0198	0.171
Money Growth (lag 1)	−0.00845	0.496
AIRI (lag 2)	−0.00777	0.604
Money Growth (lag 2)	−0.000392	0.974

contains predictive content for macroeconomic variables (Baker, Bloom, and Davis 2016; Goshima et al. 2020), and extends it to economies where timely statistics are scarcest.

4.3 Granger Causality

The Dumitrescu–Hurlin test rejects the null that AIRI does not Granger-cause inflation for the full panel (Z-bar = 3.199, p = 0.001). This is the most robust result in the paper: the test allows the relationship to be present in some countries and absent in others, and invokes no restriction on contemporaneous relationships, so it holds independently of the Cholesky ordering. The regime split shows the relationship is far from uniform. The null is rejected decisively among high-inflation economies but cannot be rejected among low-inflation ones, and the statistic in the high-inflation group exceeds that for the full panel, indicating that pooling dilutes the result rather than generating it. Two interpretations are available. The behavioural reading is that attention is state-dependent: where inflation is low and stable, agents rationally economise on attention and news carries little influence, whereas where inflation is high, beliefs are updated more frequently and news becomes a genuine input into pricing decisions, consistent with the information rigidity framework of Coibion and Gorodnichenko (2015). The measurement reading is that the index has less to work with where inflation is low, since fewer articles are published and the variance of AIRI falls, as the near-flat series for Mauritania and Tanzania suggest.

These readings are two views of one mechanism, because falling attention produces falling coverage and therefore lower index variance. The evidence cannot establish that the rejection of the null is purely behavioural. Under either reading the implication is the same and is favourable: the index is informative precisely where inflation risk is highest and conventional statistics are weakest.

Table 9: Panel Granger Causality Test Results

Sample	N	Z-bar	p-value	Sig. (5%)?
Full Panel	46	3.199	**0.001**	**Yes**
HIGH inflation	23	3.547	0.000	Yes
LOW inflation	23	0.977	0.329	No

Table 10: Impulse Response Function (IRF) Results

Horizon	Response	Std. error	95% lower	95% upper	Significant?
0	0.027	0.077	−0.123	0.177	No
1	0.287	0.091	0.109	0.464	Yes
2	0.318	0.083	0.155	0.482	Yes
3	0.401	0.118	0.169	0.634	Yes
4	0.410	0.118	0.180	0.641	Yes
5	**0.419**	0.120	0.183	0.655	Yes
6	0.415	0.117	0.185	0.645	Yes
7	0.404	0.113	0.183	0.626	Yes
8	0.389	0.108	0.178	0.600	Yes
9	0.372	0.102	0.172	0.572	Yes
10	0.355	0.097	0.166	0.544	Yes
11	0.337	0.091	0.158	0.516	Yes
12	0.319	0.087	0.149	0.488	Yes

Note: Horizon is months after the shock. "Significant?" means the 95% interval excludes zero. N = 46 countries, 2015–2025, monthly data.

4.4 Impulse Response

The impulse response function traces the dynamic effect of a shock to AIRI on inflation. The result shows that the effect on inflation starts from horizon one and peaks at 0.419 at horizon 5 and starts to decay; the result also stayed significant until horizon 12 and retains 76% of its peak magnitude, 0.319. This indicates that the AIRI produces a persistent effect on inflation. Therefore, not only does AIRI predict inflation, the magnitude of the predictive relation builds over several months. This result is consistent over the different orderings of the other four variables in the model (see Appendix Table A2).

For robustness the reverse ordering was checked, placing AIRI last, allowing it to have influence only after month 0, setting the impact response to zero. The impact of AIRI was significant from month 2 through month 12, peaking at 0.326 in horizons five and six and maintaining 80 per cent of the peak at horizon 12. This confirms that the impact of AIRI on inflation does not depend on whether AIRI is allowed to affect inflation within the month (Appendix Table A1).

4.5 Addressing the Outlier

South Sudan inflation figures were observed to be high, 95.7% average inflation, relative to the next country, Angola, which was 21.8%. However, to capture as much of the available data as possible, it was included in the model. To confirm that this outlier does not affect our conclusion, we ran the analysis without South Sudan. We found that our conclusion stays the same but the effect differed. The AIRI coefficient remains significant but the coefficient falls from 0.08 to 0.045, the Granger test still rejects the null, and the impulse response keeps its shape and stays significant through horizon 12 but with a reduced response peak from 0.42 to 0.34, and the same-month response becomes statistically significant.

4.6 Forecast Evaluation

The Clark–West test was prioritised because of the nested nature of our forecast model. Clark and West (2007) proposed that the MSPE (Mean Squared Prediction Error) adds to the forecast error of larger models because additional variables introduce noise.

Table 11 reports the two Clark–West comparisons (M1/M2 and M0/M2) and the RMSE of each model, for all 46 countries and excluding South Sudan. With all 46 countries, AIRI does not improve significantly on the macroeconomic model (M1) at any horizon, and improves on the AR baseline (M0) only at the one-month horizon. We checked whether excluding the outlier changes the outcome, given that South Sudan supplies about 75 to 80 per cent of the pooled squared forecast errors of the AR benchmark, and 90 per cent of the pooled variation in lagged inflation, so the pooled coefficients mostly reflect its own dynamics and forecast the other countries poorly.

Excluding South Sudan, the AIRI model improves significantly on both benchmarks at every horizon; the Clark–West result showed that the AIRI model is better than the baseline and macro models at every horizon, and its RMSE is lower than the baseline's by 0.3 to 0.4 per cent at one, six and twelve months.

For the robustness check, the countries were scaled by dividing their Clark–West scores by their variance and averaged (Table 12). This method gives each country comparable estimates and ensures that the forecast gains are not merely influenced by the outliers. The result showed that, when South Sudan is excluded, AIRI is better than the macro model at all horizons and better than the baseline at horizon 1.

Table 11: Out-of-Sample Forecast Evaluation

Panel A. All 46 countries

h	p	RMSE M0	RMSE M1	RMSE M2	CW M1/M2	CW M0/M2
1	12	3.264	3.770	3.781	1.39 (0.082)	**2.33 (0.010)**
3	12	5.325	5.830	5.886	0.39 (0.347)	1.00 (0.157)
6	12	8.837	11.444	11.468	0.04 (0.484)	1.08 (0.140)
12	12	14.219	15.590	15.621	−0.70 (0.758)	1.24 (0.107)

Panel B. Excluding South Sudan (N = 45)

h	p	RMSE M0	RMSE M1	RMSE M2	CW M1/M2	CW M0/M2
1	10	1.316	1.321	1.312	**4.11 (<0.001)**	**4.46 (<0.001)**
3	12	2.611	2.650	2.628	**2.84 (0.002)**	**2.18 (0.015)**
6	7	4.435	4.479	4.417	**3.27 (0.001)**	**3.33 (<0.001)**
12	12	7.933	7.977	7.905	**1.88 (0.030)**	**2.39 (0.008)**

Table 12: Equal-Weight Clark–West Statistics

h	All 46: M1/M2	All 46: M0/M2	Excl. SS: M1/M2	Excl. SS: M0/M2
1	0.55 (0.290)	0.71 (0.239)	**2.90 (0.002)**	**2.67 (0.004)**
3	0.03 (0.488)	−0.11 (0.543)	**1.83 (0.034)**	1.06 (0.145)
6	−0.55 (0.710)	−0.35 (0.636)	**2.77 (0.003)**	0.88 (0.188)
12	−2.15 (0.984)	**1.67 (0.047)**	**2.22 (0.013)**	0.60 (0.275)

5 Conclusion

The level of inflation is a critical metric that feeds into economic planning, not only because it concerns welfare but because economic expansions sponsored by investment activities largely depend on policymakers' ability to control inflation. Data availability and timeliness influence the effectiveness of policymaking. African economies are frequently exposed to inflation shocks, yet conventional inflation predictors are constrained by reporting lags and data limitations that leave policymakers reacting to price pressures only after they have already materialised.

This paper proposes an index, the African Inflation Risk Index (AIRI), that identifies inflation risk leveraging indications in news media across African countries. Desirable qualities of this index include its timeliness, transparency, and auditability. Statistical tests were carried out to investigate the predictability of the AIRI index.

The empirical results support AIRI's value as a leading indicator. The Mean Group Panel VAR, adopted specifically to accommodate the heterogeneity in African economies, found that AIRI's first lag is a statistically significant predictor of next-period inflation even after controlling for inflation's own persistence, exchange rate growth, GDP growth, and

money growth. This result is consistent with a growing literature demonstrating that news-based indicators have macroeconomic information.

Panel Granger causality tests corroborated this finding, rejecting the null hypothesis that AIRI does not Granger-cause inflation. Splitting the panel by inflation regime revealed that this predictive relationship is not uniform: it is significant in high-inflation economies, but cannot be statistically distinguished from zero in low-inflation economies. The impulse response function result also effect on inflation starts from horizon one and peaks 0.419 at horizon 5 and starts to decay, also, the result stayed significant until horizon 12 and retains 76% of its peak magnitude, 0.319.For the out-of-sample forecast evaluation, using the Clark–West test appropriate given the nested structure of the competing models, found that AIRI significantly improves inflation forecasts at horizons of up to six month.

From the statistical evidence we see that the AIRI index is a useful alternative to mainstream inflation predictors.

5.1 Summary of Findings

- The panel exhibits strong cross-sectional dependence, and all five series are stationary under both the IPS and the cross-sectionally augmented CIPS tests, so the panel VAR is estimated in levels.
- The first lag of the AIRI inflation pillar is a positive and statistically significant predictor of next-period inflation after conditioning on two lags of inflation, exchange rate depreciation, money growth and output growth. The second lag is insignificant, consistent with a signal whose informational value is concentrated in the immediately preceding month.
- The Dumitrescu–Hurlin test rejects non-causality from AIRI to inflation for the full panel. This is the paper's most robust result, since it does not depend on any identifying assumption.
- The relationship is strongly state-dependent. It is decisive among high-inflation economies and statistically indistinguishable from zero among low-inflation economies, consistent with attention-dependent expectation formation and with the reduced variance of a coverage-based index in low-inflation settings.
- The response of inflation to an AIRI shock is negligible on impact, emerges at horizon one, peaks at horizon five and remains significant through horizon twelve. The persistence reflects propagation through an inertial inflation process and is inconsistent with a purely contemporaneous reporting channel.

- Out of sample, AIRI improves on the macroeconomic benchmark at every horizon only when South Sudan is excluded, and the gains in forecast accuracy are negligible.

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Appendix

Table A1: Robustness Check: AIRI Ordered Last

Horizon	Response	Std. error	95% lower	95% upper	Significant?
0†	0.000	0.000	0.000	0.000	No
1	0.209	0.133	−0.052	0.469	No
2	0.226	0.100	0.030	0.421	Yes
3	0.302	0.136	0.034	0.569	Yes
4	0.314	0.132	0.055	0.574	Yes
5	**0.326**	0.134	0.063	0.589	Yes
6	**0.326**	0.130	0.071	0.582	Yes
7	0.321	0.126	0.074	0.569	Yes
8	0.312	0.121	0.074	0.549	Yes
9	0.300	0.116	0.073	0.527	Yes
10	0.287	0.111	0.070	0.505	Yes
11	0.274	0.106	0.066	0.482	Yes
12	0.262	0.102	0.062	0.461	Yes

Note: Same estimation and conventions as Table 10. Ordering: GDP growth, money growth, exchange-rate growth, inflation, AIRI (the exact reverse of Table 10). † Zero by construction: with AIRI ordered last, an AIRI shock cannot affect inflation within the same month. Bold marks the peak (months 5 and 6 are equal to three decimals).

Table A2: Invariance Check: Other Variables Reordered, AIRI Still First

Horizon	Response	Std. error	95% lower	95% upper	Significant?
0	0.027	0.077	−0.123	0.177	No
1	0.287	0.091	0.109	0.464	Yes
2	0.318	0.083	0.155	0.482	Yes
3	0.401	0.118	0.169	0.634	Yes
4	0.410	0.118	0.180	0.641	Yes
5	**0.419**	0.120	0.183	0.655	Yes
6	0.415	0.117	0.185	0.645	Yes
7	0.404	0.113	0.183	0.626	Yes
8	0.389	0.108	0.178	0.600	Yes
9	0.372	0.102	0.172	0.572	Yes
10	0.355	0.097	0.166	0.544	Yes
11	0.337	0.091	0.158	0.516	Yes
12	0.319	0.087	0.149	0.488	Yes

Note: Same estimation and conventions as Table 10. Ordering: AIRI, exchange-rate growth, money growth, inflation, GDP growth. Every figure is identical to Table 10.